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Lecture 03A

Before-Tutorial-Session

Tutorial 2

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HIDDEN

Theorem. (Conditions for Subspace) Let $\mathcal{V}$ be a vector space.

A subset $\mathcal{U}$ of $\mathcal{V}$ is a subspace if and only if $\mathcal{U}$ satisfies the following conditions.

- **(Zero Vector).** $\mathbf{0}$ belongs to $\mathcal{U}$.
- **(Closure under Vector Addition).** For all $u, v \in \mathcal{U}$, we have **BLANK1** $u + v \in \mathcal{U}$
- **(Closure under Scalar Multiplication).** For $\lambda \in \mathbb{F}, u \in \mathcal{U}$, we have **BLANK2** $\lambda u \in \mathcal{U}$

Fill in the blanks.

We will solve:

Question 2. Consider the following subsets $\mathcal{W} \subseteq \mathcal{P}_4(\mathbb{R})$ and $\mathcal{V} \subseteq \mathcal{P}_4(\mathbb{R})$, where

$$\mathcal{W} = \{p(x) \in \mathcal{P}_4(\mathbb{R}) \mid p(0) = 0 \text{ and } p(2) = 0\},$$

and

$$\mathcal{V} = \{p(x) \in \mathcal{P}_4(\mathbb{R}) \mid p(0) = 0 \text{ or } p(2) = 0\}.$$

- (a) Is $\mathcal{W}$ is a subspace of $\mathcal{P}_4(\mathbb{R})$?
- (b) Is $\mathcal{V}$ is a subspace of $\mathcal{P}_4(\mathbb{R})$?

Solution Plan

For $\mathcal{V}$ and $\mathcal{W}$, we **try** to show three conditions.

Consider $\mathcal{W} = \{p(x) \in \mathcal{P}_4(\mathbb{R}) : p(0) = 0 \text{ and } p(2) = 0\}$.

We want to show that $\mathcal{W}$ has the zero vector.

What is the zero vector in $\mathcal{W}$?

1. Zero matrix
- ✓ 2. Zero polynomial $\rightarrow$ aka zero function $z(x) = 0$ for all x
3. Number Zero

Consider $\mathcal{W} = \{p(x) \in \mathcal{P}_4(\mathbb{R}) : p(0) = 0 \text{ and } p(2) = 0\}$.

So, does **0** belong to $\mathcal{W}$?

↳ 0 refers to $z(x)$

For $z(x) \in \mathcal{W}$,

$z(0) = 0$ & $z(2) = 0 \Rightarrow$ because z is zero function

$\therefore$ Yes 0 belong to $\mathcal{W}$.

Consider $\mathcal{W} = \{p(x) \in \mathcal{P}_4(\mathbb{R}) : p(0) = 0 \text{ and } p(2) = 0\}$.

We want to show that $\mathcal{W}$ is closed under vector addition.

Let $f, g \in \mathcal{W}$. How do we show that $f + g \in \mathcal{W}$?

1. Show that $(f + g)(0) = 0$ ONLY.

2. Show that $(f + g)(0) = 0$ and $(f + g)(2) = 0$. $f + g \in \mathcal{W}$

3. Show that $(f + g)(0) = 0$ or $(f + g)(2) = 0$.

show $f(0) = 0$ and $f(2) = 0$ only $\Rightarrow f \in \mathcal{W}$

show $g(0) = 0$ and $g(2) = 0$ only $\Rightarrow g \in \mathcal{W}$

Consider $\mathcal{W} = \{p(x) \in \mathcal{P}_4(\mathbb{R}) : p(0) = 0 \text{ and } p(2) = 0\}$.

So, is $\mathcal{W}$ closed under $\oplus$?

For $f, g \in \mathcal{W}$.

$$(f+g)(0) = f(0) + g(0) = 0 + 0 = 0$$

because $f \in \mathcal{W}$ *because $g \in \mathcal{W}$*

Similarly can show $(f+g)(2) = 0$

$$(f+g)(0) = 0 \text{ \& } (f+g)(2) = 0$$

$$\Rightarrow f+g \in \mathcal{W}.$$

We want to show that $\mathcal{W}$ is closed under scalar multiplication.

Let $\lambda \in \mathbb{R}$ and $f \in \mathcal{W}$. How to show that $(\lambda f) \in \mathcal{W}$?

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1. Show that $(\lambda f)(0) = 0$ and $(\lambda f)(2) = 0$.
2. Show that $(\lambda f)(0) = 0$ or $(\lambda f)(2) = 0$.

Consider $\mathcal{W} = \{p(x) \in \mathcal{P}_4(\mathbb{R}) : p(0) = 0 \text{ and } p(2) = 0\}$.

So, is $\mathcal{W}$ closed under $\otimes$?

For $\lambda \in \mathbb{R}$ and $f \in \mathcal{W}$

show $\lambda f \in \mathcal{W}$ (How? D.I.Y)

Consider $\mathcal{V} = \{p(x) \in \mathcal{P}_4(\mathbb{R}) : p(0) = 0 \text{ or } p(2) = 0\}$.

Is Albert's proof that " $\mathcal{V}$ is closed under vector addition" correct?

Albert's Solution: Let $p(x) = x$ and $q(x) = 1048576x$.

Since $p(0) = 0$, we have $p(x) \in \mathcal{V}$.

Similarly, $q(0) = 0$, we have $q(x) \in \mathcal{V}$.

Then $(p + q)(x) = 1048577x$.

Since $(p + q)(0) = 0$, we have $(p + q)(x) \in \mathcal{V}$.

Therefore, $\mathcal{V}$ is closed under vector addition.

Not correct!
→ cannot prove by
example

Consider $\mathcal{V} = \{p(x) \in \mathcal{P}_4(\mathbb{R}) : p(0) = 0 \text{ or } p(2) = 0\}$.

Consider the incorrect proof that $\mathcal{V}$ is closed under vector addition.

(Step 1) Let $f, g \in \mathcal{V}$ and so,

either (A) $f(0) = g(0) = 0$, or (B) $f(2) = g(2) = 0$.

(Step 2) So, either (A) $(f + g)(0) = f(0) + g(0) = 0$, or (B) $(f + g)(2) = f(2) + g(2) = 0$

(Step 3) Therefore, $f + g$ belongs to $\mathcal{V}$.

Which of the following statements is **NOT** correct?

1. Step 1 is correct.
2. Step 1 imply Step 2.
3. Step 2 imply Step 3.

$f(x) = x$ $g(x) = x - 2$
 Does f belong to $\mathcal{V}$? Yes $f(0) = 0$
 Does g belong to $\mathcal{V}$? Yes $g(2) = 0$
 Does $f(0) = g(0)$? No. $f(0) = 0, g(0) = -2$
 Does $f(2) = g(2)$? No. $f(2) = 2, g(2) = 0$

We will solve questions similar to:

Question 6. In the following parts, you are given a real vector space $\mathcal{V}$, a vector $\mathbf{v}$ in it, and a subset S of the vector space. Determine if the given vector $\mathbf{v}$ lies in the span of the given subset S . If it does, find an explicit representation of $\mathbf{v}$ as a linear combination of the vectors in S .

(a) $\mathcal{V} = \mathbb{R}^3$, $\mathbf{v} = (5, 1, -5)$, and $S = \{(1, -2, -3), (-2, 3, -4)\}$.

(b) $\mathcal{V} = \mathcal{M}_{2 \times 2}(\mathbb{R})$, $\mathbf{v} = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$, and $S = \left\{ \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \right\}$.

(c) $\mathcal{V} = \mathcal{P}_3(\mathbb{R})$, $\mathbf{v} = 2 + 3x - 11x^2 - 2x^3$, and $S = \{-1 + 3x - 2x^2 + x^3, -2 + 3x + x^2 + 2x^3\}$.

Does $\mathbf{v}$ belong to $\text{span}(S)$

$$\begin{bmatrix} 1 & 61 \\ 80 & 33 \end{bmatrix}$$

$$v = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Set $v =$? and $S = \left\{ \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \right\}$.

To determine v in $\text{span}(S)$, what is the system of equations should you solve?

1. System 1

$$\begin{aligned} \lambda_1 &= 1 \\ \lambda_1 + \lambda_2 &= 61 \\ \lambda_1 + \lambda_2 + \lambda_3 &= 80 \\ \lambda_1 + \lambda_2 + \lambda_3 &= 33 \end{aligned}$$

2. System 2

$$\begin{aligned} \lambda_1 - \lambda_2 - \lambda_3 &= 1 \\ \lambda_1 + \lambda_2 - \lambda_3 &= 61 \\ \lambda_1 - \lambda_2 + \lambda_3 &= 80 \\ \lambda_1 + \lambda_2 + \lambda_3 &= 33 \end{aligned}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \lambda_1 \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \lambda_2 \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} + \lambda_3 \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

$$1 + 61x + 80x^2 + 33x^3$$

Set $v =$? and

$$S = \{ -1 + 3x - 2x^2 + x^3, \quad -2 + 3x + x^2 + 2x^3 \}.$$

To determine v in $\text{span}(S)$, what is the system of equations should you solve?

✓ 1. System 1

$$\begin{aligned} -\lambda_1 - 2\lambda_2 &= 1 \\ 3\lambda_1 + 3\lambda_2 &= 61 \\ -2\lambda_1 + \lambda_2 &= 80 \\ \lambda_1 + 2\lambda_2 &= 33 \end{aligned}$$

2. System 2

$$\begin{aligned} -\lambda_1 - 2\lambda_2 &= 3 \\ 3\lambda_1 + 3\lambda_2 &= 14 \\ -2\lambda_1 + \lambda_2 &= 15 \\ \lambda_1 + 2\lambda_2 &= 92 \end{aligned}$$

Let S be a subset of a vector space.

Is there a vector that always belong to $\text{span}(S)$?

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